CURRICULUM VITAE · OCTOBER 2026

# Manuel Giron

Physics · Experimental photonics · Scientific computing

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**Research interests:** high energy physics and quantum field theory; experimental optics and photonics; quantum and semiconductor physics; scientific computing.

## EDUCATION

**Florida Polytechnic University** Aug 2025 – May 2028 (expected)

Bachelor of Science in Physics, Quantum Mechanics Concentration

GPA: 3.85/4.00 · Provost Scholarship

Relevant coursework: Quantum Mechanics I, Electromagnetism I, Semiconductor Physics, Applied Partial Differential Equations; additional coursework in experimental physics, electronics, and numerical methods.

**Miami Dade College** Aug 2023 – May 2025

Associate in Arts · GPA: 4.00/4.00 · Summa Cum Laude · President's List

## RESEARCH EXPERIENCE

**Research Assistant, Experimental Photonics** May 2026 – present

Florida Polytechnic University

- Research random lasing and optical emission from biological scatterers, including seashell-derived particles in Rhodamine B–glycerol gain media; review literature on disordered optical feedback and emission narrowing.
- Prepare biological samples and optical scattering suspensions; conduct pulsed-laser excitation and collect wavelength-resolved emission spectra using laboratory spectrometers and laser-energy measurement equipment.
- Develop Python workflows to process spectroscopy datasets, identify and fit peaks, calculate linewidths and peak intensities, and compare optical emission under different experimental conditions.
- Interface spectrometers, energy meters, motorized stages, and Arduino-controlled components for automated experiments and data acquisition; create research figures and technical documentation.
- Contribute experimental interpretations and materials to research posters and scientific manuscript preparation.

**Photonic Crystal Optical Glazing** 2026

Numerical research project

- Designed and simulated a 33-layer $TiO_2/CaF_2$ optical short-wave-pass glazing for passive solar heat management in hot-humid climates, incorporating a 188 nm $Si_3N_4$ Fabry–Perot cavity.
- Implemented transfer-matrix calculations with wavelength-dependent Sellmeier refractive indices across 400–1600 nm; compared the cavity design with a layer-count-matched reference.
- Calculated an approximately 40% narrower 10–90% cut-on transition, 88.9% photopic visible transmittance, and modeled insulating-glass-unit SHGC ≈0.215. All findings are simulation-derived, without fabrication.
- Evaluated humidity durability considerations and angular dependence of near-infrared rejection; coauthored the resulting Applied Optics publication.

## PUBLICATIONS & RESEARCH PRESENTATIONS

1. Emadelden Fouad, A. F. Amin, **Manuel Giron**, and Arafa H. Aly, "Humidity-resilient $TiO_2/CaF_2$ photonic crystal smart window with a Fabry–Perot cavity for broadband near-infrared blocking in hot-humid subtropical climates: a design and simulation study," *Applied Optics* 65, 7235–7244 (2026). doi:10.1364/AO.606352.
2. Rebecca R. Robinson, **Manuel Giron**, Hengzhou Liu, and Nathan J. Dawson, "Raman mode selection in dye-infused avian feather lasers with scattering-based feedback." Manuscript in preparation.
3. Poster presenter, NanoFlorida 2026 conference.

## TECHNICAL & MODELING PROJECTS

**Mathematical Contest in Modeling (MCM/ICM)** Feb 2026

Lunar colony logistics

- Developed a quantitative logistics model for a proposed 100,000-person lunar settlement, accounting for transportation capacity, population growth, infrastructure needs, and resource constraints.
- Performed sensitivity analyses of launch and payload utilization assumptions, propulsion performance, and operational losses to evaluate logistics feasibility and comparative system performance.

**NASA MINDS · Mission 4 Methane** May 2023 – Mar 2024

Miami Dade College

- Developed a concept for in-situ methane production from Martian atmospheric feedstock to support on-site liquid-propellant supply.
- Evaluated competing methane production and purification pathways, including recycling loops, condensation, and membrane separation; identified design approaches targeting greater than 90% product purity.
- Placed third in the senior division of the NASA MINDS competition.

**Finite-Element Heat-Equation Solver** 2026

Numerical methods coursework

- Implemented a one-dimensional finite-element heat-equation model with Dirichlet boundary conditions and implicit-Euler time stepping; evaluated convergence on a manufactured solution.

## PROFESSIONAL & TEACHING EXPERIENCE

**SAT Tutor · SC Tutoring** May – Dec 2025

- Provided one-to-one SAT instruction using targeted practice and diagnostic feedback; documented an approximately 100-point score improvement for a student.
- Prepared progress summaries and communicated achievement and areas for improvement to students, parents, and supervisors.

## LEADERSHIP & SERVICE

**Phi Sigma Pi · Vice President & Founding Member** Aug 2025 – present

Helped establish Florida Polytechnic University’s chapter and support its leadership, community development, and organizational growth.

**Society of Hispanic Professional Engineers · Member** Aug 2025 – present

## TECHNICAL SKILLS & LANGUAGES

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|---|---|
| **Programming** | Python (NumPy, SciPy, Matplotlib), MATLAB, C, LabVIEW, LaTeX; numerical analysis, data visualization, spectral processing, and curve fitting. |
| **Research methods** | Optical spectroscopy, pulsed-laser experiments, experimental data acquisition, photonic-crystal transfer-matrix modeling, numerical simulations, technical literature review. |
| **Instrumentation** | Spectrometers, optical energy meters, motion-control systems, Arduino-based relays and control, circuits and PCB design, SpectraSuite. |
| **Languages** | Spanish (native), English (fluent), Japanese (beginner). |